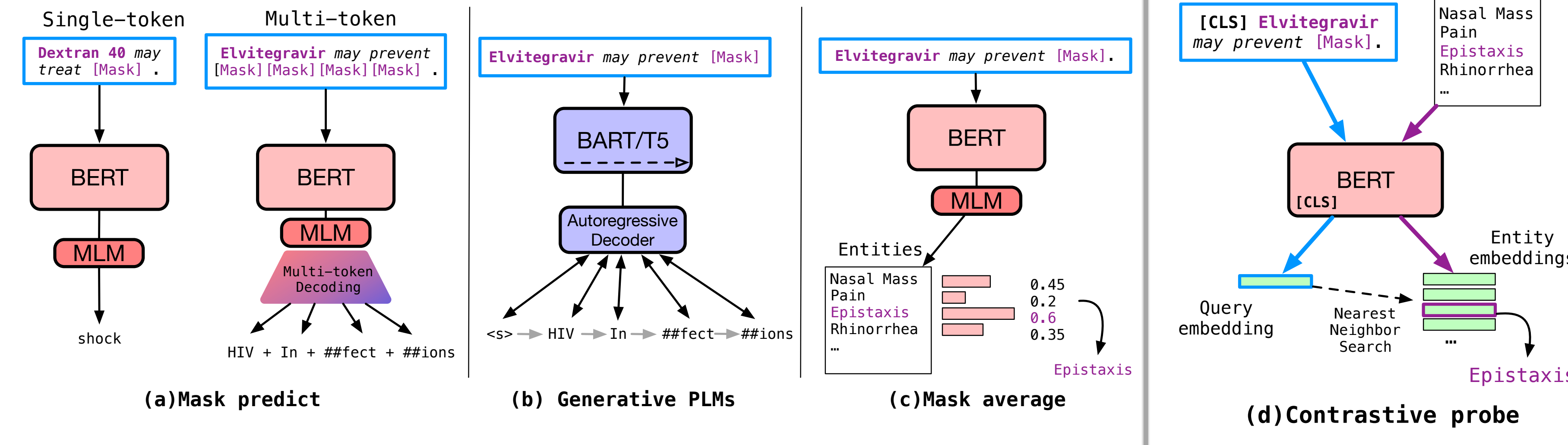


Rewire-then-Probe: A Contrastive Recipe for Probing Biomedical Knowledge of Pre-trained Language Models

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Highlights

- A new well-curated biomedical knowledge probing benchmark dataset (MedLAMA).
- A novel retrieval-based probing approach (CONTRASTIVE-PROBE) that does not require the MLM head.
- Benchmarking several state-of-the-art biomedical PLMs (including pure pre-trained and knowledge-enhanced models) on MedLAMA through our CONTRASTIVE-PROBE.



Comparison of different probing approaches.

(b) Retrieval-based Probing.

For probing step, the query is created based on the prompt-based template for each knowledge triple, as shown in the following:

Triple: Elvitegravir, may-prevent, Epistaxis
Query: Elvitegravir may prevent [MASK].

and we search for nearest neighbours from all the entity representations encoded by the same model.

Experiments

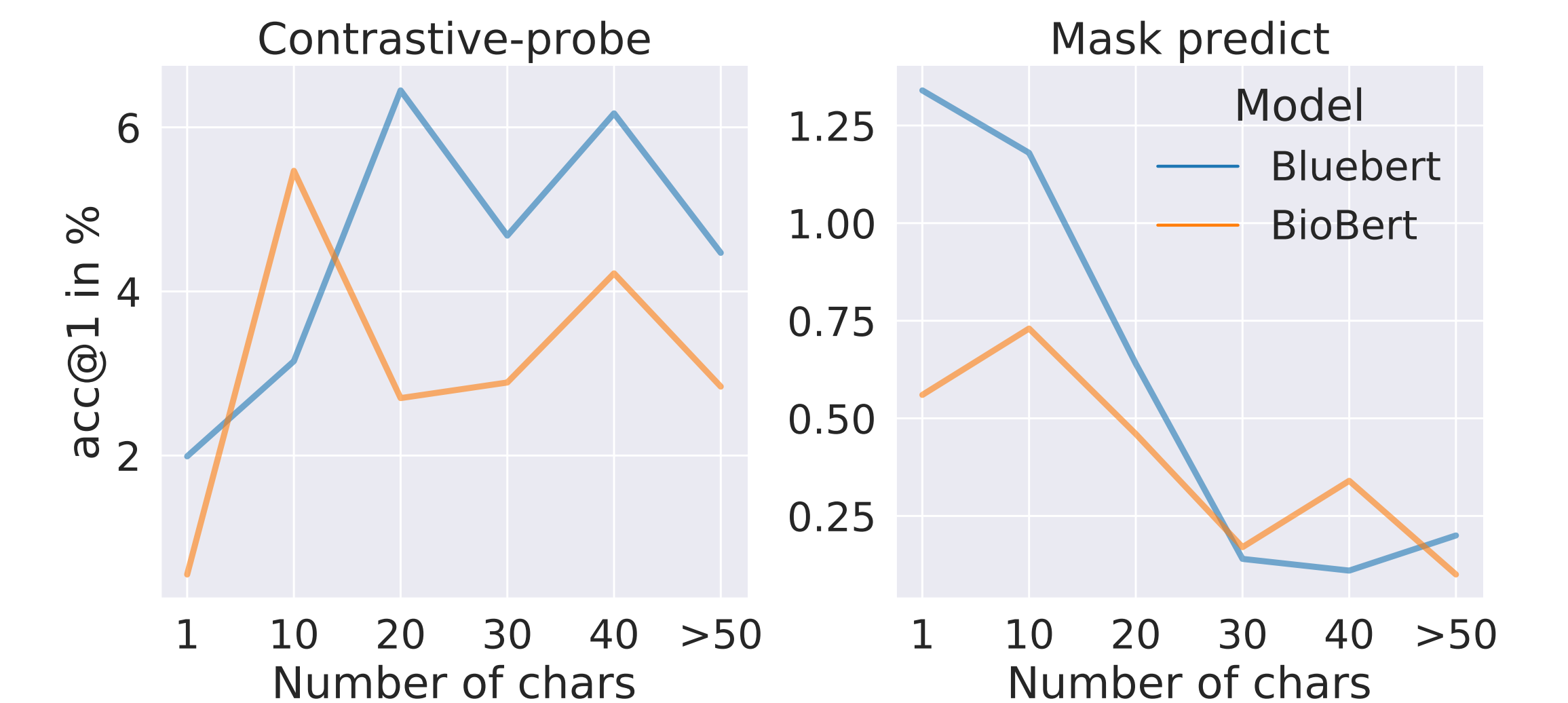
• Performance of different probing approaches on MedLAMA.

Approach	PLM	Full Set	
		acc@1	acc@10
Generative PLMs	BART-base	0.16	1.39
	SciFive-base	0.53	2.02
	SciFive-large	0.55	2.03
	T5-small	0.70	1.72
	T5-base	0.06	0.19
X-FACTR (<i>Confidence</i>)	BERT	0.05	-
	BlueBERT	0.74	-
	BioBERT	0.17	-
X-FACTR (<i>Order+Order</i>)	BERT	0.06	-
	BlueBERT	0.50	-
	BioBERT	0.11	-
Mask average	BERT	0.06	0.73
	BlueBERT	0.05	1.39
	BioBERT	0.28	3.03
Contrastive-Probe(Ours)	BERT	1.95	6.96
	BlueBERT	<u>4.87</u>	<u>19.87</u>
	BioBERT	3.28	15.46
	PubMedBERT	5.71	24.31

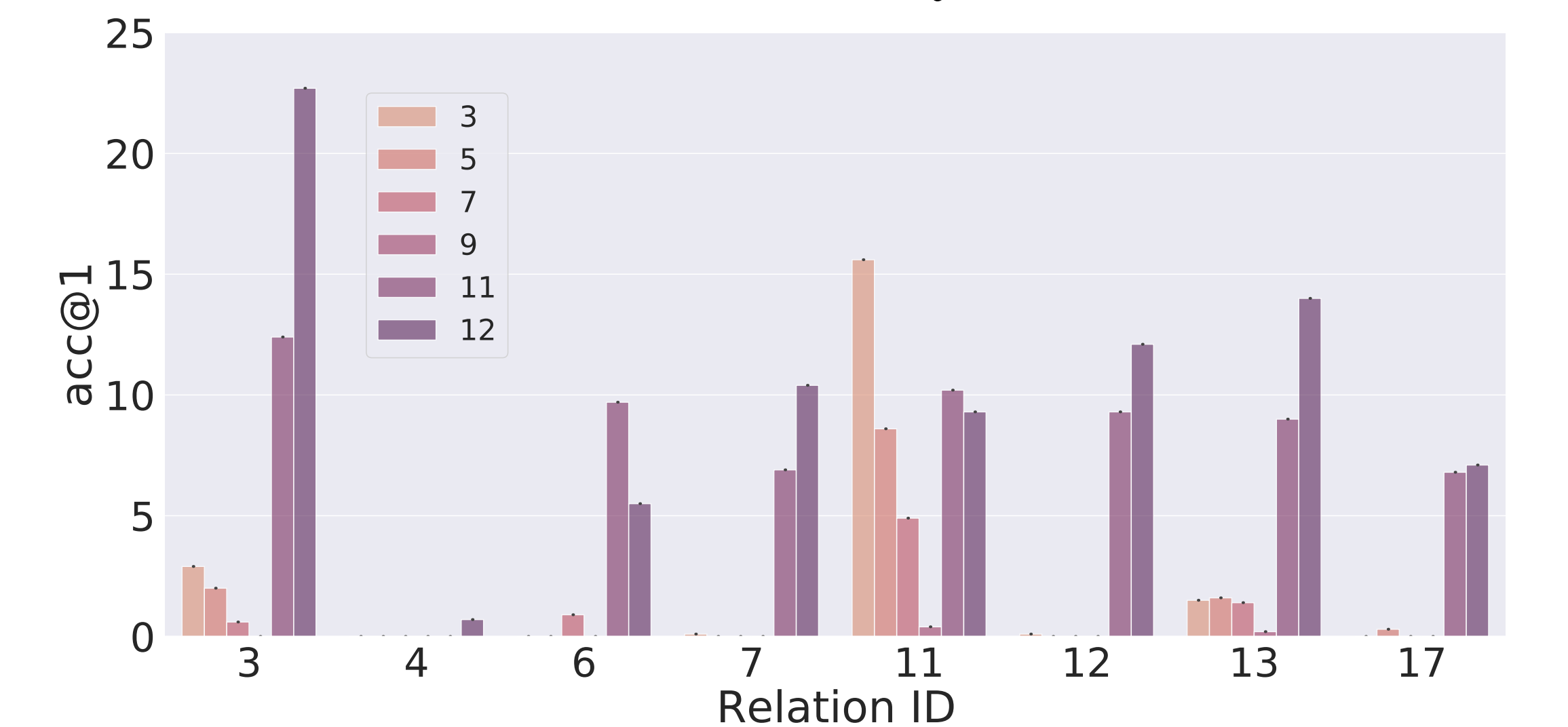
• Benchmarking biomedical PLMs via Contrastive-Probe.

Model	acc@1/acc@10	
	Full Set	Hard Set
BERT	1.95±0,40/6.96 ±0,96	0.67±0,19/3.27±0,54
BlueBERT	4.87±0,43/19.87±0,62	4.12±0,46/18.18±0,77
BioBERT	3.28±0,20/15.46±0,93	2.14±0,23/12.59±1,19
ClinicalBERT	1.83±0,15/8.64±0,79	0.71±0,13/5.45±1,06
SciBERT	3.64±0,33/18.11±1,95	2.14±0,30/14.64±2,01
PubMedBERT	5.71±0,58/24.31±1,29	4.49±0,49/21.74±1,21
UmlsBERT	2.94±0,21/11.64±0,46	1.80±0,11/7.75±0,42
SapBERT	7.80±0,38/30.41±1,23	5.15±0,27/26.09±1,17
CoderBERT	8.00±0,60/26.41±1,08	6.08±0,52/22.69±1,10

• Performance over answer lengths.



• Performance of PubMedBERT over layers and relations.



Takeaways

- Our CONTRASTIVE-PROBE is a better indicator for probing, while without using of MLM head.
- Our CONTRASTIVE-PROBE can probe layer-wise knowledge. Different layers of transformers encode different types of information, reflected by their individual success at handling certain types of prompts.
- The real lower bound of factual knowledge encoded by PLMs is higher than we estimated.
- Data and codes: <https://github.com/cambridgeltl/medlama>.

Acknowledgements

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MedLAMA

Benchmark	# Relations	# Queries	Avg. # Answer	Avg. # Char	% Single-Tokens
LAMA	41	41k	1	7.11	100 %
BioLAMA	36	49k	1	18.40	2.2 %
MedLAMA	19	19k	2.3	20.88	2.6 %

- Only 2.6 % of the answers are single-toke (Multi-token Issue).
- A significant amount of entity names are unnatural language.
- Existing probing achieves less than 1 % accuracy on MedLAMA.

Contrastive-Probe

Two steps for probing a model: a) Pre-trains the PLM on a small number of sentences sampled from the PLM's *original pre-training corpora* with a contrastive self-supervising objective; b) ask the PLM to retrieve a most relevant answer from the answer set based on the [CLS] embeddings of the query and answers.

(a) Self-supervised Contrastive Rewiring.

Sample a small set of sentences, which taken as the queries of the *cloze-style self-retrieving game* with following transformation:

Sentence: Social-distancing largely **reduces coronavirus infections**.
Query: Social-distancing largely [MASK].

where “**reduces coronavirus infections**” is marked as a positive answer of this query.

Ask PLMs to find the correct answer by optimising an InfoNCE objective. This objective function encourages PLMs to create similar representations for any query-answer pairs from the same sentence and dissimilar representations for queries/answers belonging to different sentences.